

SMART FIRE / SMOKE WARNING SYSTEM

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System Idea

INPUT SENSORS:

1. S= Smoke sensor
2. T=Temperature sensor
3. R=Reset Buton
4. M=Manual Test Switch

OUTPUTS:

1. W = Warning (YellowLed/ Slow Buzzer)
2. A = Alarm (Red Led, Fast Buzzer)
3. E = Evacuate (Continuous Buzzer+Blinking Red)

LOGIC CONCEPT:

STATUS	GREEN	YELLOW	RED	BUZZER
Safe	ON	×	×	Silent
Warning	×	ON	×	Slow
Alarm	×	×	ON	Fast
Evacuate	×	×	Flashing	Continuous
Reset	ON	×	×	Silent

WARNING

$W = S \text{ OR } T$

EVACUATE

$E = (S \text{ AND } T)$
AND
NOT R

ALARM

$A = A \text{ AND } T$



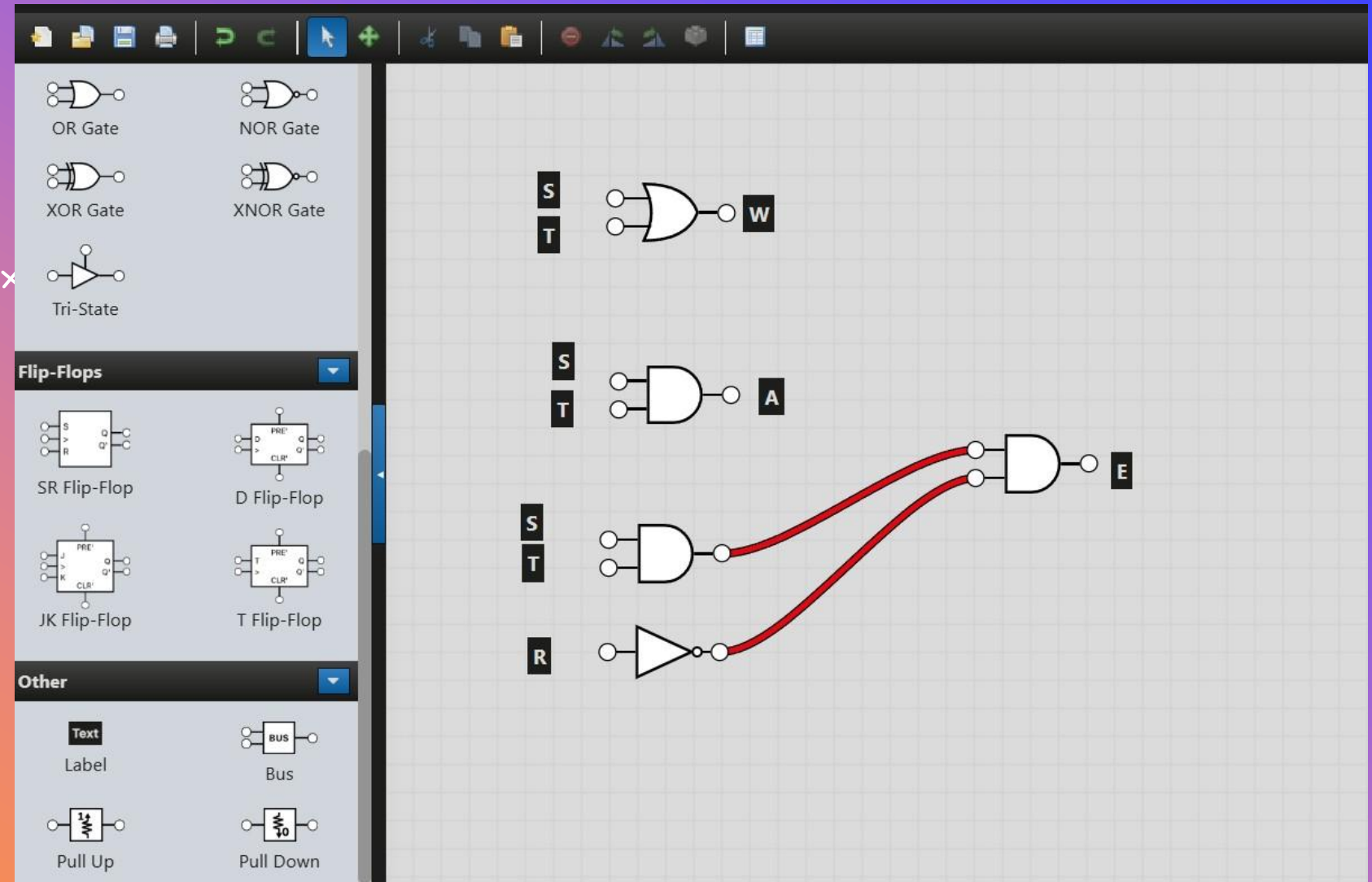
Logic Gate

TRUTH TABLE

W = OR A = AND E = (AND + NOT R)

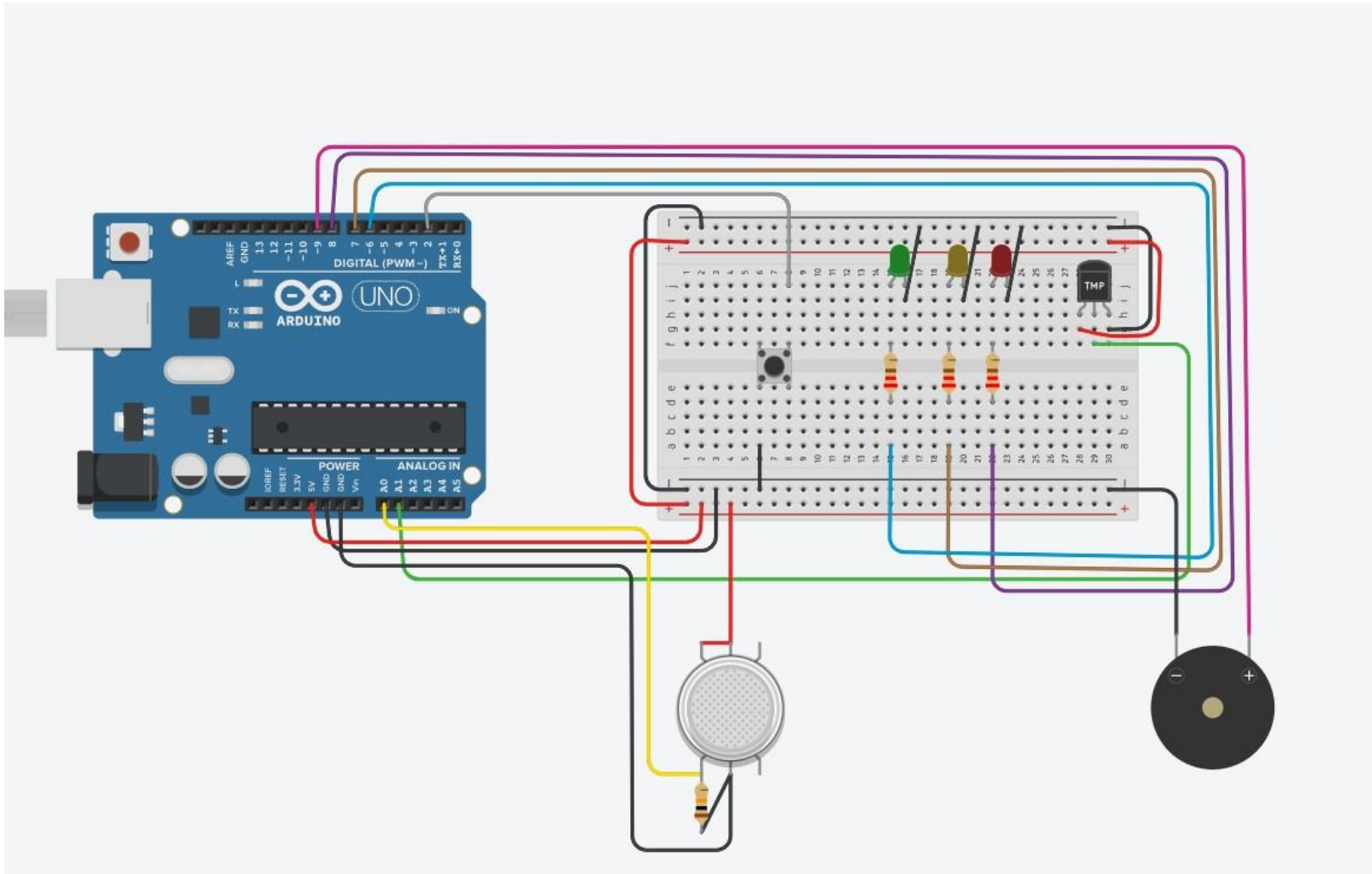
S	T	R	W	A	E
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	1	0

LOGIC GATES DIAGRAM



LOGIC.LY

TINKERCAD DESIGN

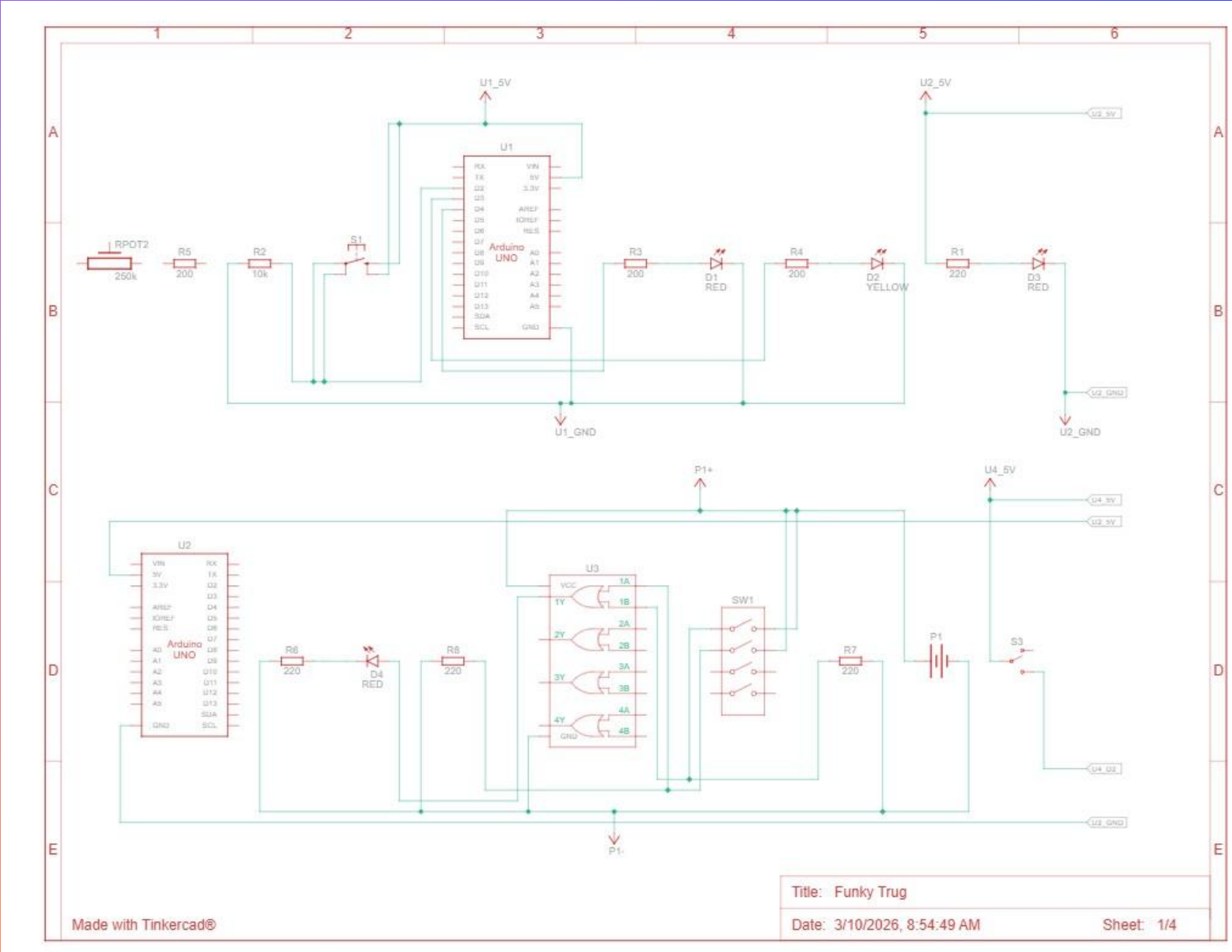


https://www.tinkercad.com/things/4fJy21QMP8D/editel?returnTo=%2Fdashboard&sharecode=hoFBjsEa2PsuOEFNwhZwsNTVHu7_IZqmhv1quBh3Kqw

SCHEMATIC VIEW

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TINKERCAD

C++

Code

Start Simulation

Send To

Text



5 (Arduino Uno R3)

```
1 int gasPin    = A0;
2 int tempPin   = A1;
3 int resetPin  = 2;
4
5 int greenLED  = 6;
6 int yellowLED = 7;
7 int redLED    = 8;
8 int buzzer    = 9;
9
10 unsigned long alarmStart = 0;
11 bool timerOn = false;
12
13 void setup() {
14
15   pinMode(greenLED, OUTPUT);
16   pinMode(yellowLED, OUTPUT);
17   pinMode(redLED, OUTPUT);
18   pinMode(buzzer, OUTPUT);
19
20   pinMode(resetPin, INPUT_PULLUP);
21
22   Serial.begin(9600);
23
24   digitalWrite(greenLED, HIGH);
25   digitalWrite(yellowLED, LOW);
26   digitalWrite(redLED, LOW);
27   digitalWrite(buzzer, LOW);
28 }
29
30 void allOFF(){
31   digitalWrite(greenLED, LOW);
32   digitalWrite(yellowLED, LOW);
33   digitalWrite(redLED, LOW);
34   digitalWrite(buzzer, LOW);
35 }
36
37 void loop() {
38
39   int gasVal = analogRead(gasPin);
40
41   float voltage = analogRead(tempPin) * (5.0 / 1023.0);
42   float tempC = (voltage - 0.5) * 100.0;
43
44   bool resetPressed = (digitalRead(resetPin) == LOW);
```

```
45
46   Serial.print("Gas: ");
47   Serial.print(gasVal);
48   Serial.print(" Temp: ");
49   Serial.println(tempC);
50
51   // RESET BUTTON
52   if(resetPressed){
53
54     allOFF();
55     timerOn=false;
56     alarmStart=0;
57
58     digitalWrite(greenLED, HIGH);
59
60     Serial.println("RESET -> SAFE");
61
62     delay(500);
63     return;
64   }
65
66   // EVACUATE CONDITION
67   if(gasVal > 700 && tempC > 60){
68
69     if(!timerOn){
70       timerOn=true;
71       alarmStart=millis();
72     }
73
74     if(millis() - alarmStart > 10000){
75
76       allOFF();
77
78       digitalWrite(buzzer,HIGH);
79
80       digitalWrite(redLED,HIGH);
81       delay(200);
```

```
82     digitalWrite(redLED,LOW);
83     delay(200);
84
85     Serial.println("EVACUATE");
86
87     return;
88   }
89
90   allOFF();
91
92   digitalWrite(redLED,HIGH);
93
94   digitalWrite(buzzer,HIGH);
95   delay(120);
96   digitalWrite(buzzer,LOW);
97   delay(120);
98
99   Serial.println("ALARM");
100
101   return;
102 }
103
104 timerOn=false;
105 alarmStart=0;
106
107 // WARNING CONDITION
108 if(gasVal > 500 || tempC > 35){
109
110   allOFF();
111
112   digitalWrite(yellowLED,HIGH);
113
114   digitalWrite(buzzer,HIGH);
115   delay(600);
116   digitalWrite(buzzer,LOW);
117   delay(600);
118
119   Serial.println("WARNING");
120
121   return;
122 }
123
124 // SAFE CONDITION
125 allOFF();
126
127 digitalWrite(greenLED,HIGH);
128
129 Serial.println("SAFE");
130
131 delay(500);
132 }
```

Activate Windows
Go to Settings to activate Windows.

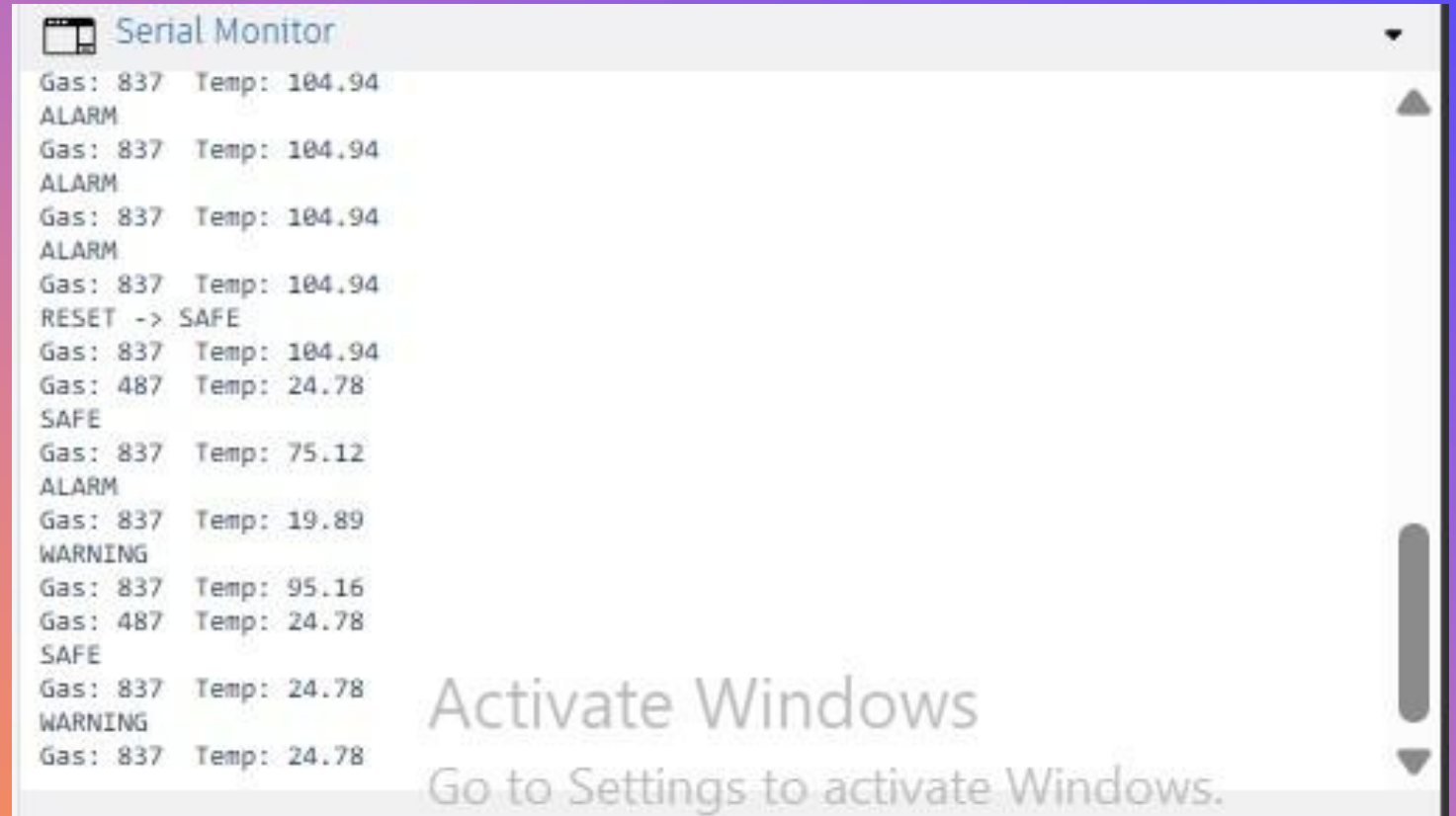
TINKERCAD

SERIAL MONITOR VIEW

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References for Smoke and Fire Alarm System Project

1. Journal / Research Papers

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2. Promise Elechi, “Design and Implementation of an Improved Smoke Detection System.” European Journal of Electrical Engineering and Computer Science, 2017.
3. Mohamad Syafiq Mohamad Adenan et al., “Smart Smoke Detector.” International Journal of Recent Technology and Applied Science, 2021.
4. Sreejith S. P., Kuldeep Baban Vayadande, “Advanced Fire Monitoring System.” International Journal of Trend in Scientific Research and Development, 2020.
5. Robert S. Allison et al., “Sensors for Fire and Smoke Monitoring.” Sensors Journal, 2021

**I hope you understand
everything clearly.**



Thank you

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